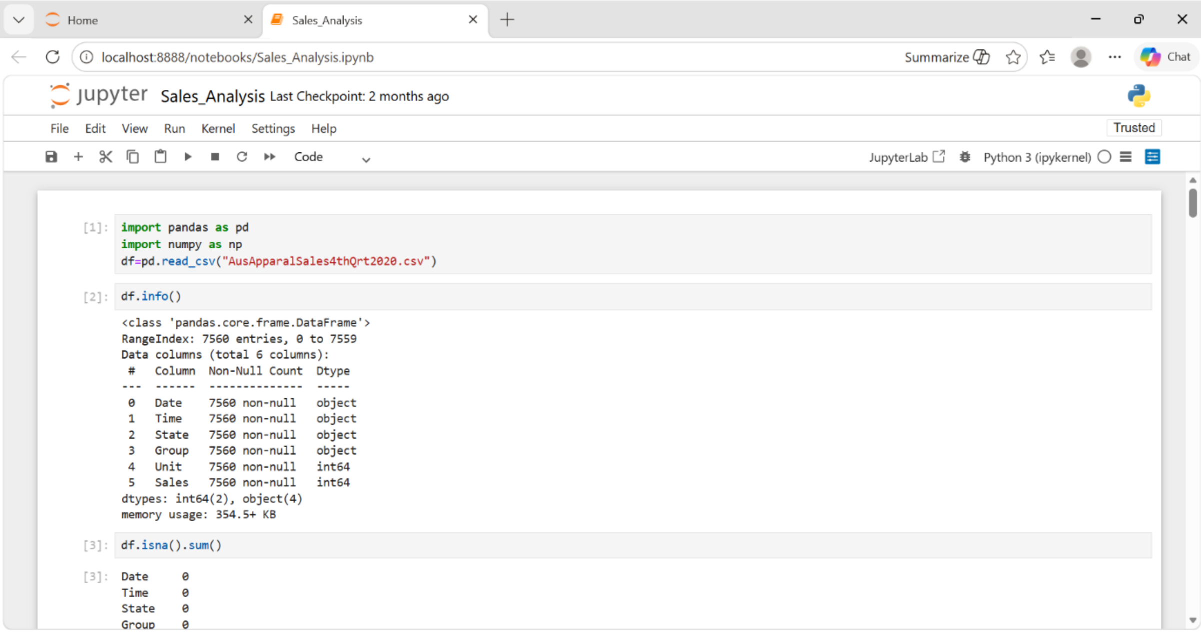
OUTPUT SCREENSHOTS



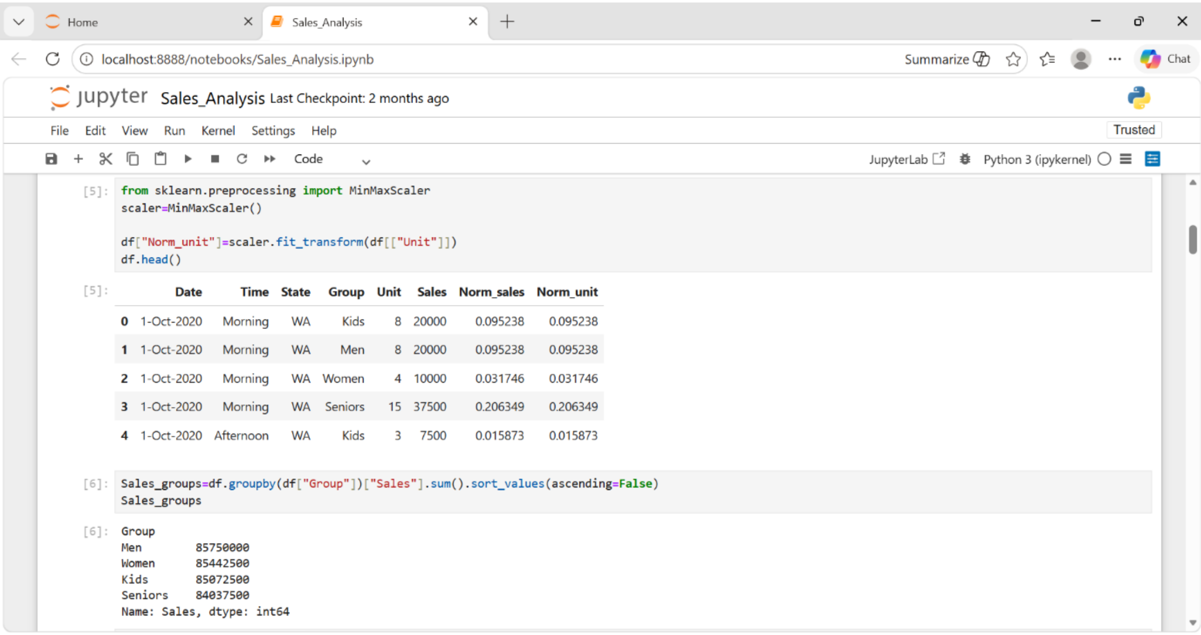

```
[1]: import pandas as pd
     import numpy as np
     df=pd.read_csv("AusApparalSales4thQrt2020.csv")

[2]: df.info()

     <class 'pandas.core.frame.DataFrame'>
     RangeIndex: 7560 entries, 0 to 7559
     Data columns (total 6 columns):
      #   Column  Non-Null Count  Dtype
     ---  ------  --------------  -----
      0   Date    7560 non-null   object
      1   Time    7560 non-null   object
      2   State   7560 non-null   object
      3   Group   7560 non-null   object
      4   Unit    7560 non-null   int64
      5   Sales   7560 non-null   int64
     dtypes: int64(2), object(4)
     memory usage: 354.5+ KB

[3]: df.isna().sum()

[3]: Date     0
     Time     0
     State    0
     Group    0
```




```
[5]: from sklearn.preprocessing import MinMaxScaler
     scaler=MinMaxScaler()

     df["Norm_unit"]=scaler.fit_transform(df[["Unit"]])
     df.head()
```

[5]:

	Date	Time	State	Group	Unit	Sales	Norm_sales	Norm_unit
0	1-Oct-2020	Morning	WA	Kids	8	20000	0.095238	0.095238
1	1-Oct-2020	Morning	WA	Men	8	20000	0.095238	0.095238
2	1-Oct-2020	Morning	WA	Women	4	10000	0.031746	0.031746
3	1-Oct-2020	Morning	WA	Seniors	15	37500	0.206349	0.206349
4	1-Oct-2020	Afternoon	WA	Kids	3	7500	0.015873	0.015873

```
[6]: Sales_groups=df.groupby(df["Group"])["Sales"].sum().sort_values(ascending=False)
     Sales_groups

[6]: Group
     Men        85750000
     Women      85442500
     Kids       85072500
     Seniors    84037500
     Name: Sales, dtype: int64
```

[7]:
```
Group
Men        34300
Women      34177
Kids       34029
Seniors    33615
Name: Unit, dtype: int64
```

[8]:
```python
buys_time=df.groupby(df["Time"])["Unit"].sum().sort_values(ascending=False)
buys_time
```

[8]:
```
Time
Morning      45683
Afternoon    45603
Evening      44835
Name: Unit, dtype: int64
```

[9]:
```python
buys_time_sales=df.groupby(df["Time"])["Sales"].sum().sort_values(ascending=False)
buys_time_sales
```

[9]:
```
Time
Morning      114207500
Afternoon    114007500
Evening      112087500
Name: Sales, dtype: int64
```

[10]:
```python
state_sales=df.groupby(df["State"])["Sales"].sum().sort_values(ascending=False)
state_sales
```

[10]:
```
State
VIC    105565000
NSW     74970000
SA      58857500
QLD     33417500
TAS     22760000
NT      22580000
WA      22152500
Name: Sales, dtype: int64
```

[11]:
```python
state_units=df.groupby(df["State"])["Sales"].sum().sort_values(ascending=False)
state_units
```

[11]:
```
State
VIC    105565000
NSW     74970000
SA      58857500
QLD     33417500
TAS     22760000
NT      22580000
WA      22152500
Name: Sales, dtype: int64
```

[12]:
```python
df["Unit"].describe()
```

[12]:
```
count    7560.000000
mean       18.005423
std        12.901403
```

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[16]:
```python
max_sales = df["Sales"].max()
min_sales = df["Sales"].min()

print("Highest Sales Value:", max_sales)
print("Lowest Sales Value:", min_sales)
```

```
Highest Sales Value: 162500
Lowest Sales Value: 5000
```

[17]:
```python
df.columns = df.columns.str.strip()
```

[18]:
```python
print(df.columns.tolist())
```

```
['Date', 'Time', 'State', 'Group', 'Unit', 'Sales', 'Norm_sales', 'Norm_unit']
```

[19]:
```python
# Clean column names (safe step)
df.columns = df.columns.str.strip()

# Convert to datetime
df["Date"] = pd.to_datetime(df["Date"])

# Set index (IMPORTANT: pass column name, not df["Date"])
df.set_index("Date", inplace=True)

print(df.head())
```

```
                  Time State     Group  Unit  Sales  Norm_sales  Norm_unit
Date
```


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[13]:
```python
df["Sales"].describe()
```

[13]:
```
count      7560.000000
mean      45013.558201
std       32253.506944
min        5000.000000
25%       20000.000000
50%       35000.000000
75%       65000.000000
max      162500.000000
Name: Sales, dtype: float64
```

[14]:
```python
from scipy import stats

mode_sales = stats.mode(df["Sales"], keepdims=True)
mode_units = stats.mode(df["Unit"], keepdims=True)
print("Mode Sales:", mode_sales.mode[0])
print("Mode Units:", mode_units.mode[0])
```

```
Mode Sales: 22500
Mode Units: 9
```

[15]:
```python
highes_group= Sales_groups.idxmax()
print("highes_group :",highes_group)
lowest_group=Sales_groups.idxmin()
print("lowest_group :",lowest_group)
```

```
highes_group :  Men
lowest_group :  Seniors
```

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```
                Time State    Group  Unit  Sales  Norm_sales  Norm_unit
Date
2020-10-01    Morning    WA     Kids     8  20000    0.095238   0.095238
2020-10-01    Morning    WA      Men     8  20000    0.095238   0.095238
2020-10-01    Morning    WA    Women     4  10000    0.031746   0.031746
2020-10-01    Morning    WA  Seniors    15  37500    0.206349   0.206349
2020-10-01  Afternoon    WA     Kids     3   7500    0.015873   0.015873
```

[20]:

```python
weekly_sales = df["Sales"].resample("W").sum()
print(weekly_sales.head())
```

```
Date
2020-10-04    15045000
2020-10-11    27002500
2020-10-18    26640000
2020-10-25    26815000
2020-11-01    21807500
Freq: W-SUN, Name: Sales, dtype: int64
```

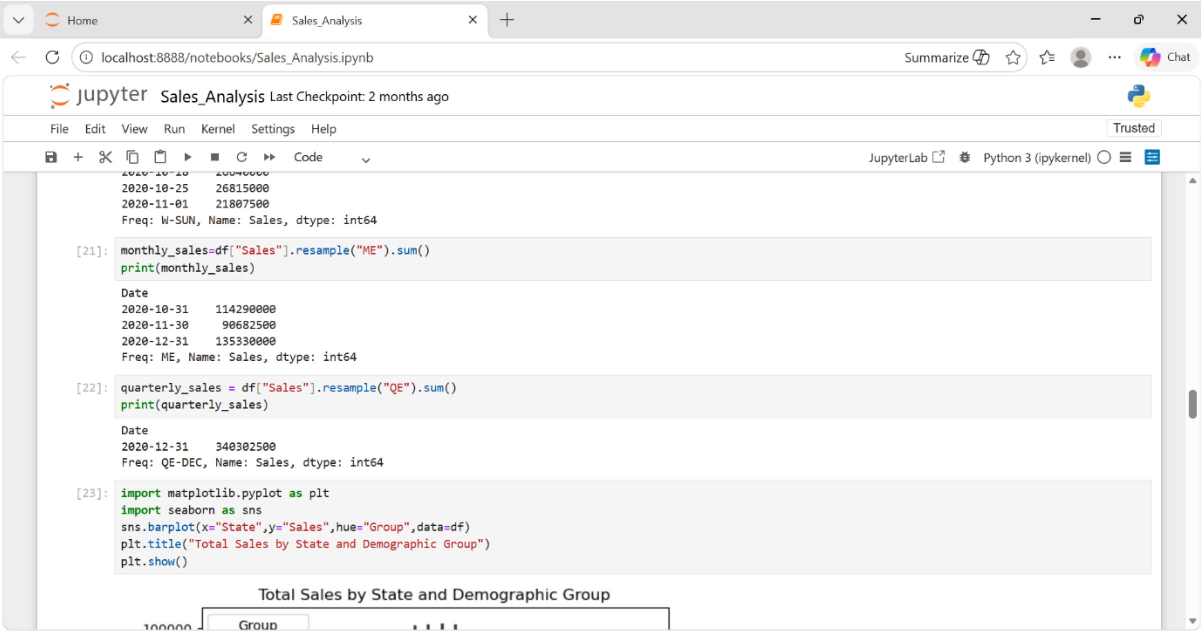
[21]:

```python
monthly_sales=df["Sales"].resample("ME").sum()
print(monthly_sales)
```

```
Date
2020-10-31    114290000
2020-11-30     90682500
2020-12-31    135330000
Freq: ME, Name: Sales, dtype: int64
```

[22]:

```python
quarterly_sales = df["Sales"].resample("QE").sum()
```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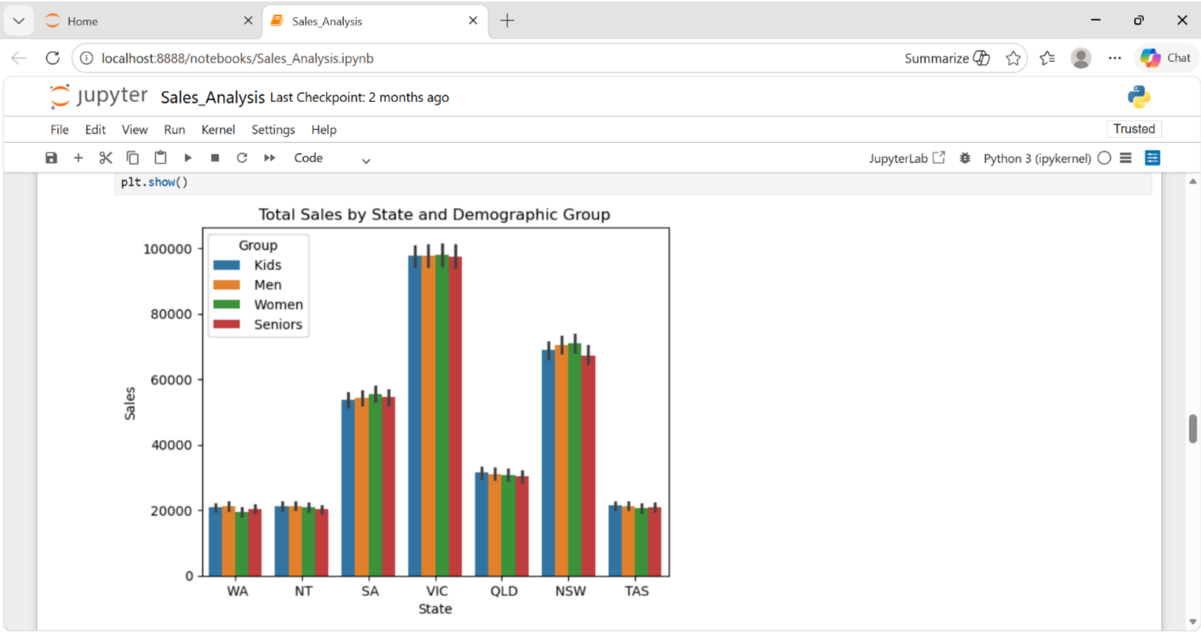
Code
JupyterLab
Python 3 (ipykernel)
2020-10-25 26815000
2020-11-01 21807500
Freq: W-SUN, Name: Sales, dtype: int64
[21]: monthly_sales=df["Sales"].resample("ME").sum()
print(monthly_sales)
Date
2020-10-31 114290000
2020-11-30 90682500
2020-12-31 135330000
Freq: ME, Name: Sales, dtype: int64
[22]: quarterly_sales = df["Sales"].resample("QE").sum()
print(quarterly_sales)
Date
2020-12-31 340302500
Freq: QE-DEC, Name: Sales, dtype: int64
[23]: import matplotlib.pyplot as plt
import seaborn as sns
sns.barplot(x="State",y="Sales",hue="Group",data=df)
plt.title("Total Sales by State and Demographic Group")
plt.show()
Total Sales by State and Demographic Group
Group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Code
JupyterLab
Python 3 (ipykernel)
plt.show()
Total Sales by State and Demographic Group
Group
Kids
Men
Women
Seniors
Sales
100000
80000
60000
40000
20000
0
WA
NT
SA
VIC
QLD
NSW
TAS
State

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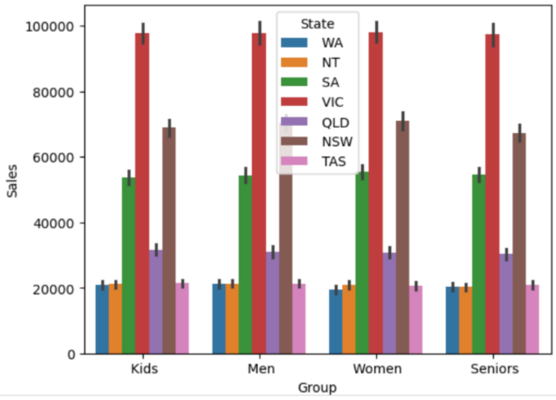
Code JupyterLab Python 3 (ipykernel)

[24]:
```
sns.barplot(x="Group",y="Sales",hue="State",data=df)
plt.show()
```



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Code JupyterLab Python 3 (ipykernel)

[29]:
```
time_sales = df.groupby("Time")["Sales"].sum().reset_index()
print(time_sales)
```

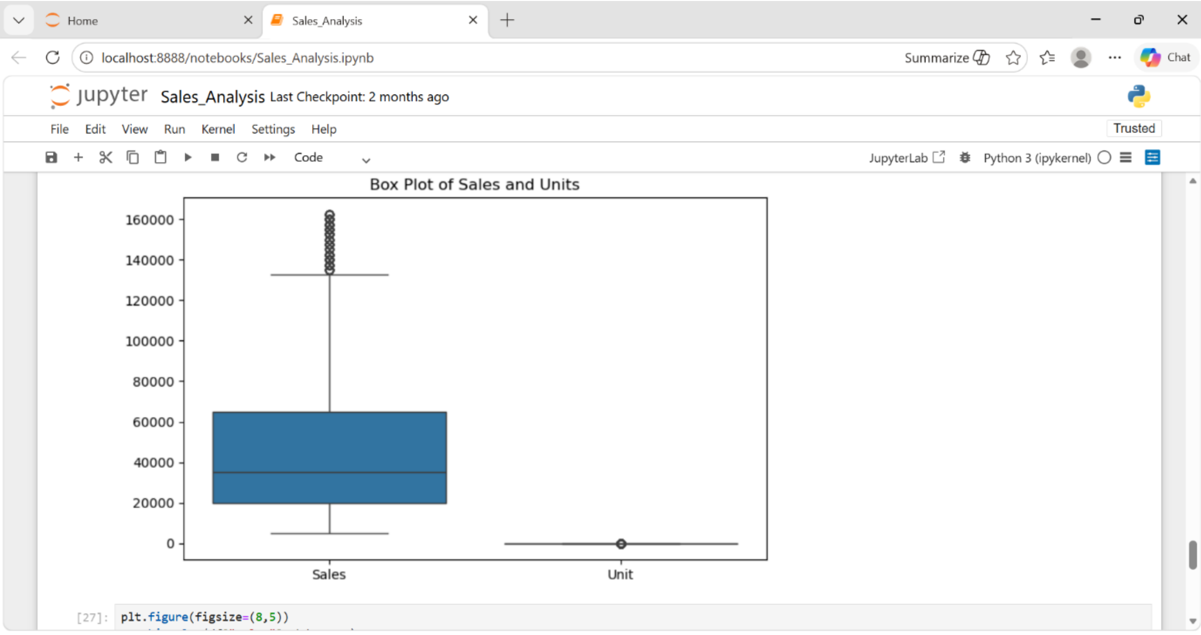
```
        Time      Sales
0  Afternoon  114007500
1    Evening  112087500
2    Morning  114207500
```

[30]:
```
peak=time_sales.loc[time_sales["Sales"].idxmax()]
off_peak=time_sales.loc[time_sales["Sales"].idxmin()]
print("peak :",peak)
print("off_peak :",off_peak)
```

```
peak : Time        Morning
Sales    114207500
Name: 2, dtype: object
off_peak : Time        Evening
Sales    112087500
Name: 1, dtype: object
```

[31]:
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
sns.barplot(data=time_sales, x="Time", y="Sales")
plt.title("Sales by Time of the Day")
plt.xlabel("Time")
```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Code
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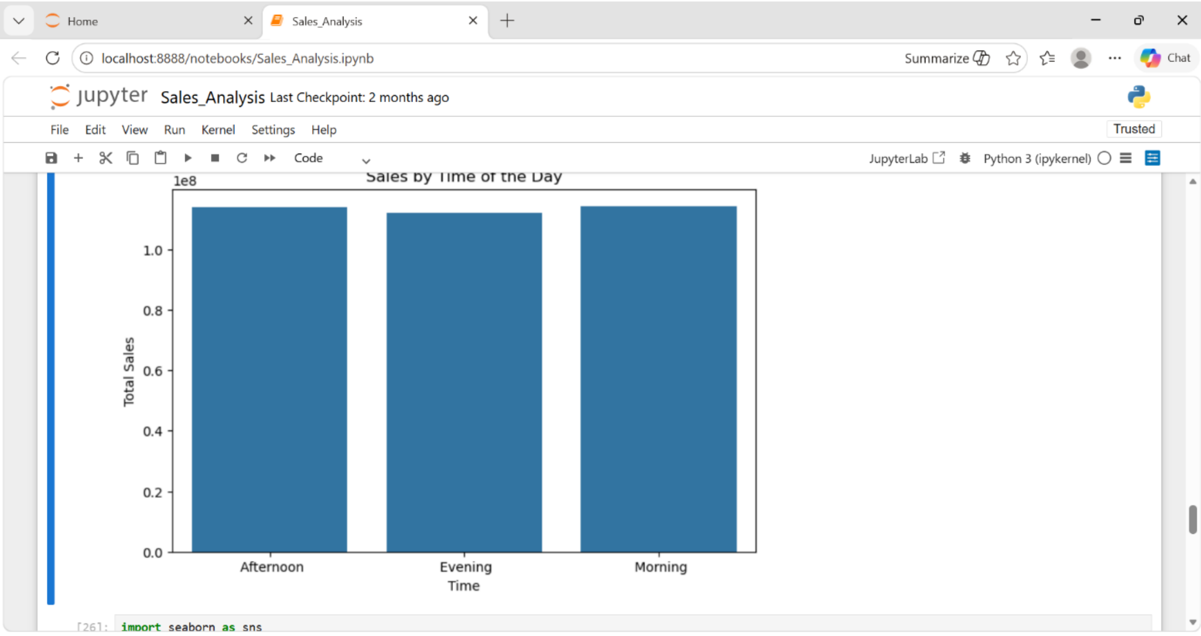
Python 3 (ipykernel)
Box Plot of Sales and Units
160000
140000
120000
100000
80000
60000
40000
20000
0
Sales
Unit
[27]: plt.figure(figsize=(8,5))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Code
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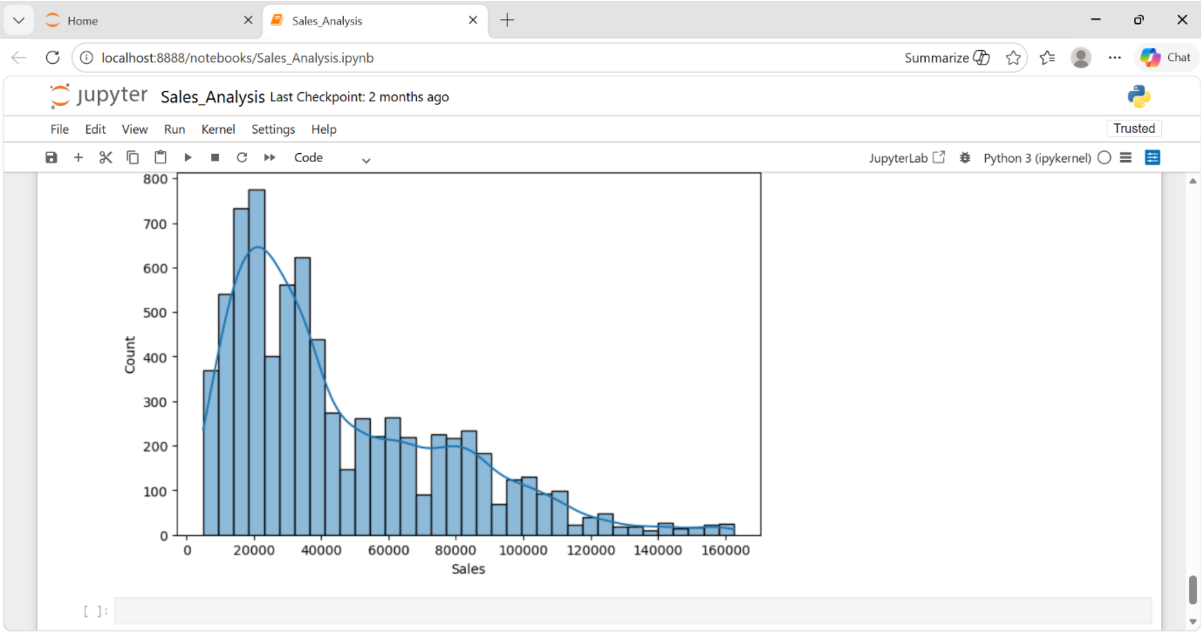
Python 3 (ipykernel)
Sales by Time of the Day
1e8
1.0
0.8
0.6
0.4
0.2
0.0
Total Sales
Afternoon
Evening
Morning
Time
[26]: import seaborn as sns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800
700
600
500
400
300
200
100
0
Count
0
20000
40000
60000
80000
100000
120000
140000
160000
Sales
[ ]: